Mohammadreza Zarifi

Email: mohammadrezazrf@gmail.com Linkedin: linkedin.com/mohammad-reza-zarifi Scholar: scholar.google.com

Education

- **Sharif University of Technology** Tehran, Iran

  M.Sc. in Mechanical Engineering; GPA: 18.82/20 (4.0/4.0) *2022 - 2025*

  Thesis Title: Development of a capsule endoscope for tissue biopsy

  Supervisor: Prof. Hossein Nejat Pishkenari

- **K.N.Toosi University of Technology** Tehran, Iran

  B.Sc. in Mechanical Engineering; GPA: 18.61/20 (3.93/4.0) *2018 - 2022*

  Project Title: Designing a Cascade Controller For a DC Motor in Presence of Disturbances

Areas of Interest & Selected Courses

- Robotics, Mechatronics, Control
- Computer Vision, Deep/Machine Learning
- **Selected Courses**: Robotic Surgery, Nonlinear Control, Artificial Neural Networks, Computer Vision, Social and Cognitive Robotics, Measurment Systems and Control

Publications

- **Journals**

  Mohammad Reza Zarifi Khameneh, Mahdi Jamali, Hossein Nejat Pishkenari
  Symmetry-based Independent Fully Convolutional Neural Networks for Magnetic Localization Using Sensor Array
  IEEE Sensors Journal 10.1109/JSEN.2025.3603679

  Mahdi Jamali, Mohammad Reza Zarifi Khameneh, Mohammad Hossein Sarkhosh, Mohammad Ali Bijarchi, Hossein Nejat Pishkenari
  Independent Position Control of Two Ferrofluid Marble Robots in Shared and Parallel Workspaces
  In Preparation

  Mahdi Jamali, Mohammad Hossein Sarkhosh, Mohammad Reza Zarifi Khameneh, Mohammad Ali Bijarchi, Hossein Nejat Pishkenari
  Adaptive Observer-Based Model Predictive Control for Magnetically Actuated Liquid Marbles
  In Preparation

- **Conferences**

  M. R. Zarifi Khameneh, Z.Tolouei, M. Yousefi, H. Nejat Pishkenari, A. Alasty
  Development and Experimental Evaluation of an Endoscopic Capsule Robot Equipped With a Fail-safe Biopsy Mechanism
  12th RSI International Conference on Robotics and Mechatronics (ICRoM 2024), 10.1109/ICRoM64545.2024.10903657

  M. Jamali, M. R. Zarifi Khameneh, M. H. Sarkhosh, H. Nejat Pishkenari
  Enhanced Optimal Control of a Magnetically Actuated Capsule Endoscope Using Electromagnetic Coils Considering Small Intestine Dynamics
  12th RSI International Conference on Robotics and Mechatronics (ICRoM 2024), 10.1109/ICRoM64545.2024.10903534

  D. Nourbakhsh Sabet, M. R. Zarifi, J. Khoramdel, Y. Borhani, E. Najafi
  An Automated Visual Defect Segmentation for Flat Steel Surface Using Deep Neural Networks
  12th International Conference on Computer and Knowledge Engineering (ICCKE2022), 10.1109/ICCKE57176.2022.9959998

Skills Summary

- **Programming Languages**: Python, C++
- **Engineering Softwares**: MATLAB & Simulink, SolidWorks & CATIA, COMSOL Multiphysics
- **Hands-on**: Raspberry Pi & Arduino, Prototyping
- **Machine Vision & Machine Learning**: Classical MV & ML, Object Detection, Image Segmentation, CNNs, LSTM/GRU

Other Projects

- **Fabricating a micromixer for mixing Rhodamine B in water**: Lab on A Chip Course Project
- **Ball and obstacle game in C++ using graphics.h library**: Computer Programming Course Project
- **A State-to-Image Approach For Isolated Sign Language Recognition**: Social and Cognitive Robotics Course Project
- **Helmet and bike detection using Yolo v4**: Artificial Neural Networks Course Project
- **Dynamic Analysis and Simulation of Knee with A Six-Bar Linkage**: Theory of Machines Course Project

Academic Experience

- **Research Assistant** — Sharif University of Technology
  Micro/Nano Robotics Lab; Director: Prof. H. Nejat Pishkenari — *Sep 2022 - Present*
- **Teaching Assistant**
  Dynamics; Instructor: Prof. H. Nejat Pishkenari — *Sep 2024 - Feb 2025*
- **Teaching Assistant**
  Measurment Systems and Control Lab; Instructor: Dr. A. Taheri — *Feb 2023 - Feb 2025*
- **Teaching Assistant**
  Applied Machine Learning; Instructor: Dr. S. Khodaygan — *Sep 2023 - Feb 2024*
- **Teaching Assistant** — K.N.Toosi University of Technology
  Mechanics of Materials II; Instructor: Dr. A. Jalali Aghchai — *Feb 2022 - Jul 2022*
- **Teaching Assistant**
  Machine Element Design I; Instructor: Dr. E. Najafi — *Feb 2022 - Jul 2022*
- **Teaching Assistant**
  Computer Programming (C++); Instructor: Dr. M. Sharbatdar — *Sep 2021 - Jul 2022*

Work Experience

- **Sharif University of Technology Patent Office** — Feb 2024 - Sep 2024
  Patent examiner: I was an examiner evaluating patents related to medical device developments.
- **Can Home Appliances Ltd.** — Jul 2021 - Sep 2021
  R&D intern: I was responsible for CAD modeling of kitchen property parts using SolidWorks.

Honors and Awards

- M.Sc. Thesis Grant Recipient: Received a research grant from the Vice President for Research and Technology of Sharif University of Technology (Grant No. AI 4030515), resulting in a peer-reviewed journal publication.
- Ranked 7th among 61 M.Sc. students of Mechanical Engineering at Sharif University of Technology - Class of 2025
- Accepted for direct M.Sc. program (without nationwide entrance exam) at Sharif University of Technology as an exceptional talent, the most prestigious university of Iran, Mechanical Engineering - 2022
- Ranked 4th among 106 B.Sc. students of Mechanical Engineering at K.N.Toosi University - Class of 2022
- Ranked in the top 2% nationwide among 134,183 participants in the National University Entrance Examination (Konkour).– 2018

Language Tests

- **English**: TOEFL IBT: 5.5/6 (110/120)

Reading:5.5 Listening:6.0 Speaking:5.0 Writing:5.5

Volunteer Experience

- **Member of Mechanical Engineering Scientific Association** — K.N.Toosi University
  Organized events, held mechanical engineering software courses for undergraduate students — *2021 - 2022*
- **Member of Theatre Society** — K.N.Toosi University
  Organized events, held workshops related to acting and playwriting — *2019 - 2021*

References

- **Hossein Nejat Pishkenari**
  Professor at Sharif University of Technology
  Email: nejat@sharif.edu
- **Alireza Taheri**
  Associate Professor at Sharif University of Technology
  Email: artaheri@sharif.edu
- **Mahkame Sharbatdar**
  Assistant Professor at K.N.Toosi University of Technology
  Email: m.sharbatdar@kntu.ac.ir